Changwoon Choi

Ph.D. Candidate, Electrical and Computer Engineering
Seoul National University

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Research Interests: 3D vision and graphics; 3D reconstruction; neuro-symbolic shape and world abstraction

Education

Seoul National University	M.S. & Ph.D., Electrical and Computer Engineering Advisor: Prof. Young Min Kim	Sep 2020 – Present
Seoul National University	B.S., Electrical and Computer Engineering	Mar 2014 – Aug 2020
Seoul Science High School		Mar 2011 – Feb 2014

Research Experience

Research Scientist Intern	Meta Reality Labs, London, UK	Jun 2025 – Dec 2025
Research Intern	NAVER Cloud, Seoul, Korea	Dec 2023 – Mar 2024
Visiting Researcher	UC San Diego, hosted by Prof. Hao Su	May 2023 – Dec 2023

Honors and Awards

Google PhD Fellowship (Machine Perception)	2026 – 2027
Outstanding Reviewer Award, CVPR 2026	2026
Meshy PhD Fellowship Finalist	2026
Outstanding Poster Presentation Award, IPIU	2025
Young Researcher Award, SNU INMC	2025
Brain Korea 21 Ph.D. Fellowship, NRF	Spring 2024
Ph.D. Study-Abroad Program Scholarship, SNU OIA ($10,000)	Spring 2023
Outstanding Teaching Assistant Award, Seoul National University	Spring 2022
Samsung Humantech Paper Award, Honorable Mention ($2,000)	2021
Merit-Based Scholarship (Full Funding), Seoul National University	2015 – 2020

Publications

*Equal contribution †Co-corresponding author

[18] Image-Guided Geometric Stylization of 3D Meshes
Changwoon Choi*, Hyunsoo Lee*, Clément Jambon, Yael Vinker, and Young Min Kim. CVPR, 2026

[17] Scene Retargeting: Learning Object Placement with Analogical Transfer
Minkwan Kim, Junho Kim, Seungmin Lee, **Changwoon Choi**, and Young Min Kim. arXiv, 2026

[16] Geometry-Aware Scene Configurations for Novel View Synthesis
Minkwan Kim, **Changwoon Choi**, and Young Min Kim. IEEE TVCG, 2026

[15] ExCellGen: Fast, Controllable, Photorealistic 3D Scene Generation from a Single Real-World Exemplar
Clément Jambon, **Changwoon Choi**, Dongsu Zhang, Olga Sorkine-Hornung, and Young Min Kim.
Computational Visual Media, 2026 – *Best Paper Honorable Mention*

[14] Continuous-time 4D Reconstruction via Event-guided Latent Feature Interpolation
Gwangtak Bae, Seunggu Kang, Junho Kim, Eun Sun Lee, **Changwoon Choi**, and Young Min Kim. BMVC, 2026

[13] Humans as a Calibration Pattern: Dynamic 3D Scene Reconstruction from Unsynchronized and Uncalibrated Videos
Changwoon Choi, Jeongjun Kim, Geonho Cha, Minkwan Kim, Dongyoon Wee, and Young Min Kim. ICCV, 2025

[12] Recovering Dynamic 3D Sketches from Videos
Jaeah Lee, **Changwoon Choi**, Young Min Kim, and Jaesik Park. CVPR, 2025

[11] 3Doodle: Compact Abstraction of Objects with 3D Strokes
Changwoon Choi, Jaeah Lee, Jaesik Park, and Young Min Kim. SIGGRAPH (ACM TOG), 2024

[10] I^2-SLAM: Inverting Imaging Process for Robust Photorealistic Dense SLAM
Gwangtak Bae*, **Changwoon Choi***, Hyeongjun Heo, Sang Min Kim, and Young Min Kim. ECCV, 2024

[9] Investigating Chiral Morphogenesis of Gold Using Generative Cellular Automata
Sang Won Im*, Dongsu Zhang*, Jeong Hyun Han, Ryeong Myeong Kim, **Changwoon Choi**, Young Min Kim†, and Ki Tae Nam†. Nature Materials, 2024

[8] Balanced Spherical Grid for Egocentric View Synthesis
Changwoon Choi, Sang Min Kim, and Young Min Kim. CVPR, 2023

[7] IBL-NeRF: Image-Based Lighting Formulation of Neural Radiance Fields
Changwoon Choi*, Juhyeon Kim*, and Young Min Kim. Pacific Graphics (Computer Graphics Forum), 2023

[6] Color Transform Module for Robust Novel View Synthesis
Sang Min Kim, **Changwoon Choi**, Hyeongjun Heo, and Young Min Kim.
Pacific Graphics (Computer Graphics Forum), 2023

[5] LDL: Line Distance Functions for Panoramic Localization
Junho Kim, **Changwoon Choi**, Hojun Jang, and Young Min Kim. ICCV, 2023

[4] Probabilistic Implicit Scene Completion
Dongsu Zhang, **Changwoon Choi**, Inbum Park, and Young Min Kim. ICLR, 2022 – *Spotlight*

[3] CPO: Change Robust Panorama to Point Cloud Localization
Junho Kim, Hojun Jang, **Changwoon Choi**, and Young Min Kim. ECCV, 2022

[2] PICCOLO: Point Cloud-Centric Omnidirectional Localization
Junho Kim, **Changwoon Choi**, Hojun Jang, and Young Min Kim. ICCV, 2021

[1] Learning to Generate 3D Shapes with Generative Cellular Automata
Dongsu Zhang, **Changwoon Choi**, Jeonghwan Kim, and Young Min Kim. ICLR, 2021

Invited Talks

"Image-Guided Geometric Stylization of 3D Meshes," Korea Computer Graphics Society	Jul 2026
"Abstraction and Style in 3D Shapes," Disney Research|Studios	Nov 2025
"Computer Vision, Computer Graphics, and AI," Seoul Science High School	Apr 2025, Apr 2026
"3Doodle: Compact Abstraction of Objects with 3D Strokes," Korea Computer Graphics Society	Jul 2024
"Towards Robust Indoor Visual Localization with Omnidirectional Camera," 42dot	May 2022

Academic Service

Conference Reviewer	SIGGRAPH, SIGGRAPH Asia, Pacific Graphics, NeurIPS, CVPR, ICCV, ECCV, 3DV, WACV, ACCV, BMVC, CAD/Graphics
Journal Reviewer	ACM TOG, IEEE TVCG, Computer Graphics Forum, Pattern Recognition

Patents

US 12,664,720; KR 10-2813248; KR 10-2969785

Teaching

Teaching Assistant	Physics Lab; Introduction to Random Variables and Processes; Signals and Systems; Theory and Lab of IoT, AI, and Big Data	2019 – 2022
	3D Computer Vision Track for AI Experts (Samsung)	Summer 2021, 2022
Tutor	Signals and Systems; Engineering Mathematics II; Programming Methodology; Calculus I, II; Physics I, II	2015 – 2021